Question No:- 2

Give short answer of the following:

Draw $(0\bar{1}0)$, (020), (632) and (010) crystallographic planes in cubic unit cell.

$(0\bar{1}0)$

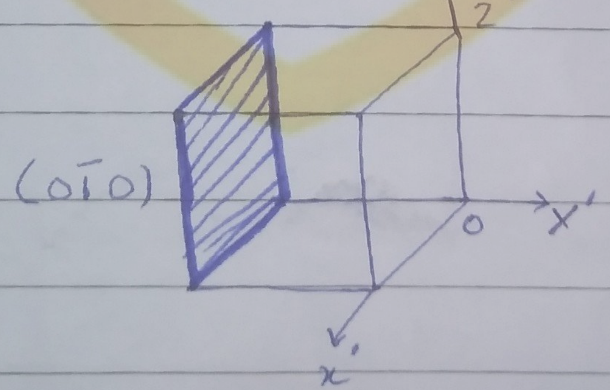
The intercepts on x, y, z axis are $(\infty, -1, \infty)$

By taking reciprocal we get

$(0, -1, 0)$

$h, k, l = 0, -1, 0$

$(hkl) = (0\bar{1}0)$



$(0,2\,0)$

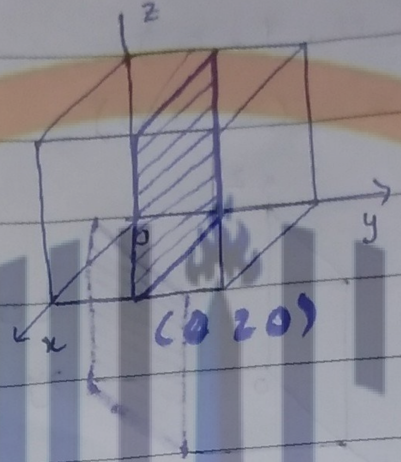
The intercepts on x, y, z axis are $(\infty, \frac{1}{2}, \infty)$

Day: Date:

By taking resiprocal we get

(0, 2, 0)

(h k l) = (0 2 0)



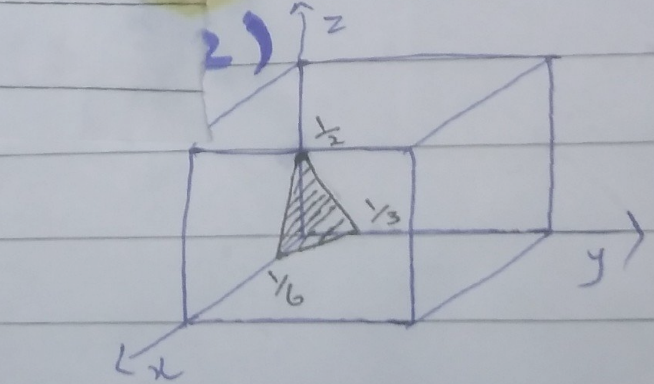
(6 3 2)

The intercepts on x, y, z axis are $\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$

by Taking reciprocal we get

$\cancel{\left(1, \frac{1}{2}, \frac{1}{3}\right)}$ (6, 3, 2)

(h k l) = (6 3 2)



Day: ________ Date: ________

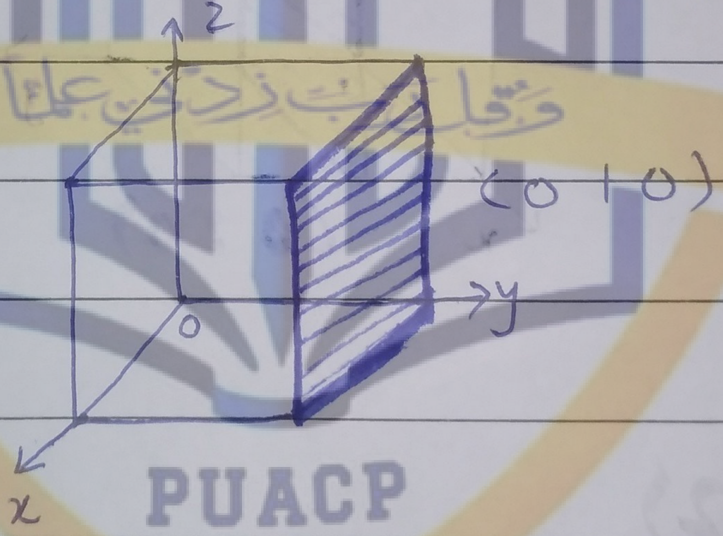
(0 1 0)

The intercepts on x, y, z axis are $(\infty, 1, \infty)$

By taking inverse we get

$(0, 1, 0)$.

$(h\ k\ l) = (0\ 1\ 0)$.



(b)

What is meant by quantization of elastic waves? Discuss briefly.

polyatomic lattices. There is a frequency gap at the boundary $K_{max} = \pm\pi/a$ of the first Brillouin zone.

QUANTIZATON OF ELASTIC WAVES

The energy of a lattice vibration is quantized. The quantum of energy is called a **phonon** in analogy with the photon of the electromagnetic wave. The energy of an elastic mode of angular frequency ω is

$$\epsilon = (n + \tfrac{1}{2})\hbar\omega \tag{27}$$

when the mode is excited to quantum number n; that is, when the mode is occupied by n phonons. The term $\frac{1}{2}\hbar\omega$ is the zero point energy of the mode. It occurs for both phonons and photons as a consequence of their equivalence to a quantum harmonic oscillator of frequency ω, for which the energy eigenvalues are also $(n + \frac{1}{2})\hbar\omega$. The quantum theory of phonons is developed in Appendix C.

We can quantize the mean square phonon amplitude. Consider the standing wave mode of amplitude

$$u = u_0 \cos Kx \cos \omega t \ .$$

Here u is the displacement of a volume element from its equilibrium position at x in the crystal. The energy in the mode, as in any harmonic oscillator, is half kinetic energy and half potential energy, when averaged over time. The kinetic energy density is $\frac{1}{2}\rho(\partial u/\partial t)^2$, where ρ is the mass density. In a crystal of volume V, the volume integral of the kinetic energy is $\frac{1}{4}\rho V\omega^2 u_0^2 \sin^2 \omega t$. The time average kinetic energy is

$$\tfrac{1}{8}\rho V\omega^2 u_0^2 = \tfrac{1}{2}(n + \tfrac{1}{2})\hbar\omega \ , \tag{28}$$

because $\langle\sin^2 \omega t\rangle = \frac{1}{2}$. The square of the amplitude of the mode is

$$u_0^2 = 4(n + \tfrac{1}{2})\hbar/\rho V\omega \ . \tag{29}$$

This relates the displacement in a given mode to the phonon occupancy n of the mode.

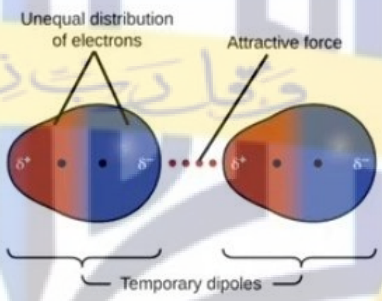
What is the sign of ω? The equations of motion such as (2) are equations for ω^2, and if this is positive then ω can have either sign, + or −. But the

short question 2

energy of a phonon must be positive, so it is conventional and suitable to view ω as positive. If the crystal structure is unstable, then ω^2 will be negative and ω will be imaginary.

Van der Waals-London Interaction

The interaction occurs due to the <u>DIPOLAR INTERACTION</u> between the two neighboring atoms, causing weak attractive force between the atoms knowns as dispersion bonds. The dipole thus produced is not permanent but keep oscillating with movement of electronic cloud around the nucleus. It is non directional in nature.



short question no 3

Van der Waals-London Interaction

- Present in all molecules but most prominent in inert gases.
- Since these forces are weak, thus the bond can break at room temperature and these forces become ineffective.
- That's why these molecules usually exist in gaseous form.

Want to see more:

https://www.youtube.com/watch?v=T73I30fsAc8&feature=emb_logo

Mathematical Formalism (Vander Waals)

As a model, we consider two identical linear harmonic oscillators 1 and 2 separated by R. Each oscillator bears charges $\pm e$ with separations x_1 and x_2

$$K.E. = \frac{P_2^2}{2m}$$

$$P.E. = \frac{1}{2} m\omega^2 x_2^2$$

$$K.E. = \frac{P_1^2}{2m}$$

$$P.E. = \frac{1}{2} m\omega^2 x_1^2$$

$$H_o = \frac{P_1^2}{2m} + \frac{1}{2} m\omega^2 x_1^2 + \frac{P_2^2}{2m} + \frac{1}{2} m\omega^2 x_2^2$$

Continue..

H_o is the Hamiltonian of the unperturbed system, and a constant 'C' can be introduced i.e., $C = m\,\omega_o^2$.

Consider H_1 as the coulomb interaction energy of the two oscillators:

$$\text{(CGS)} \qquad \mathcal{H}_1 = \frac{e^2}{R} + \frac{e^2}{R + x_1 - x_2} - \frac{e^2}{R + x_1} - \frac{e^2}{R - x_2} ; \qquad (2)$$

$$\Delta U = \tfrac{1}{2}\hbar(\Delta\omega_s + \Delta\omega_a) = -\hbar\omega_0 \cdot \frac{1}{8}\left(\frac{2e^2}{CR^3}\right)^2 = -\frac{A}{R^6} . \qquad (9)$$

Continue..

$$\Delta U = -\frac{A}{R^6}$$

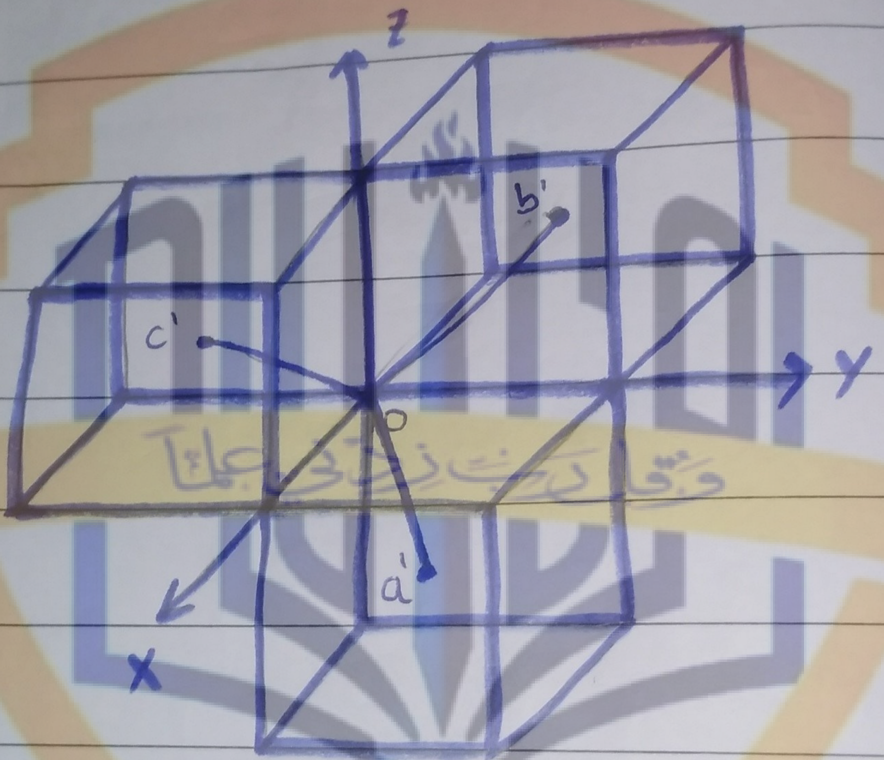
Thus, interaction varies as the minus 6th power of the separation between two oscillators.

For two identical atoms, approximate value of 'A' is $\hbar\omega_o\alpha^2$ where $E = \hbar\omega_o$ and α is the electronic polarizability.

- This is called the van der Waals interaction/ London interaction or the induced dipole-dipole interaction.
- It is the principal attractive interaction in crystals of inert gases and in the crystals of many organic molecules.

short question 4

Show that reciprocal of BCC lattice is a FCC lattice.



a', b' and c' are the primittive translation vectors of BCC crystal lattice. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the orthognal unit vectors along x, y and z axis respectively.

from fig we get

$$\vec{a}' = \frac{a}{2}\hat{i} + \frac{a}{2}\hat{j} - \frac{a}{2}\hat{k}$$

$$\vec{b}' = -\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}$$

$$\vec{c}' = \frac{a}{2}\hat{i} - \frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}$$

let $\vec{a}^*$, $\vec{b}^*$ and $\vec{c}^*$ be the primittive translation vectors of resiprocal lattice.

$$\vec{a}^* = \frac{2\pi\,(\vec{b}' \times \vec{c}')}{\vec{a}' \cdot (\vec{b}' \times \vec{c}')}$$

$$\vec{b}^* = \frac{2\pi\,(\vec{c}' \times \vec{a}')}{\vec{a}' \cdot (\vec{b}' \times \vec{c}')}$$

$$\vec{c}^* = \frac{2\pi\,(\vec{a}' \times \vec{b}')}{\vec{a}' \cdot (\vec{b}' \times \vec{c}')}$$

$$\vec{b}' \times \vec{c}' = \begin{vmatrix} i & j & k \\ -\frac{a}{2} & \frac{a}{2} & \frac{a}{2} \\ \frac{a}{2} & -\frac{a}{2} & \frac{a}{2} \end{vmatrix}$$

$$= i\left(\frac{a^2}{4} + \frac{a^2}{4}\right) - j\left(-\frac{a^2}{4} - \frac{a^2}{4}\right) + k\left(\frac{a^2}{4} - \frac{a^2}{4}\right)$$

$$= \frac{2a^2}{4} i + \frac{2a^2}{4} j = \frac{a^2}{2}(i+j)$$

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Similarly we can find

$(\vec{c}' \times \vec{a}') = \frac{a^2}{2}(\hat{j}+\hat{k})$

$(\vec{a}' \times \vec{b}') = \frac{a^2}{2}(\hat{k}+\hat{i})$

$\vec{a}' \cdot (\vec{b}' \times \vec{c}') = \left(\frac{a}{2}i + \frac{a}{2}j - \frac{a}{2}k\right) \cdot \left(\frac{a^2}{2}(\hat{i}+\hat{j})\right)$

$= \frac{a^3}{4} + \frac{a^3}{4}$

$= \frac{a^3}{2}$

Put these value in a^* b^* and c^*

$$a^* = \frac{2\pi \frac{a^2}{2}(\hat{i}+\hat{j})}{\frac{a^3}{2}}$$

$$\boxed{a^* = \frac{2\pi}{a}(\hat{i}+\hat{j})}$$

$$b^* = \frac{2\pi \frac{a^2}{2}(\hat{j}+\hat{k})}{\frac{a^3}{2}}$$

$$\boxed{b^* = \frac{2\pi}{a}(\hat{j}+\hat{k})}$$

$$c^* = \frac{2\pi \left(\frac{a^2}{2}(\hat{k}+\hat{i}\right)}{\frac{a^3}{2}}$$

$$\boxed{c^* = \frac{2\pi}{a}(\hat{k}+\hat{i})}$$

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These reciprocal lattice vectors of BCC are the primitive translation vectors of FCC Lattice. Thus resiprocal lattice of BCC is FCC lattice.

Calculate the packing fraction of HCP structure.

short question 5

Definition:-

"The ratio between filled and unfilled or the ratio between occupied and non occupied space in a unit cell is called packing fraction."

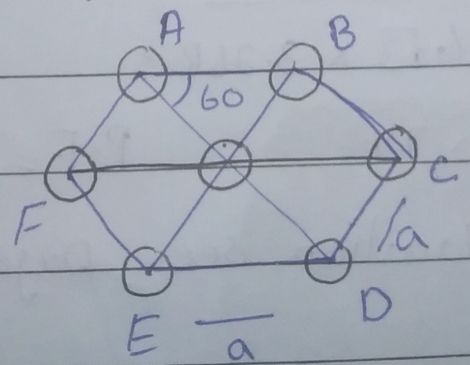
$$P.F = \frac{n \times \text{volume of one atom}}{\text{volume of unit cell.}}$$

Packing fraction of HCP:-

In hexagonal structure $n=6$

volum of one atom $= \frac{4}{3}\pi r^3$

volume of unit cell $=$ base area $\times$ height



The area of equilateral triangle

$OAB = \frac{1}{2} a \times a \sin 60$

$= \frac{\sqrt{3}}{4} a^2$

Now, area of base plane

$= 6 \times \frac{\sqrt{3}}{4} a^2$

$= \frac{3\sqrt{3}}{2} a^2$

Since $a = 2R$

So area of base plane is

$= \frac{3\sqrt{3}}{2} (2R)^2$

$= 6\sqrt{3} R^2$

Since $\frac{c}{a} = 1.633$

$\therefore$

$c = 1.633 (2R)$

$c = 3.26R$

$$P.F = \frac{6 \times \frac{4}{3} \pi R^3}{6\sqrt{3} \times 3.26R^3}$$

$= 0.74$ $\qquad$ $P.F = 74\%$

"For best explanation see. Page No 20 from notes"

zone ... but inside $+\frac{2\pi}{a}$ is known as 2^{nd} brillouin

The region outside the 2^{nd} brillouin zone but inside $+\frac{3\pi}{a}$ is known as 3^{rd} brillouin zone and so on.....

long question 1

Dynamics of a Diatomic Linear Lattice

In this section we shall consider slightly more complicated case, the lattice dynamics of a diatomic chain having masses M_1 and M_2 where($M_1>M_2$) and the nearest neighbor distance is 'a'. This is one dimensional analogue of three dimensional Nacl structure. For the convenience, the atoms are numbered in such a way that the even numbers correspond M_1 and the odd numbers to M_2. The following figure shows the equilibrium and displaced positions of such a system.

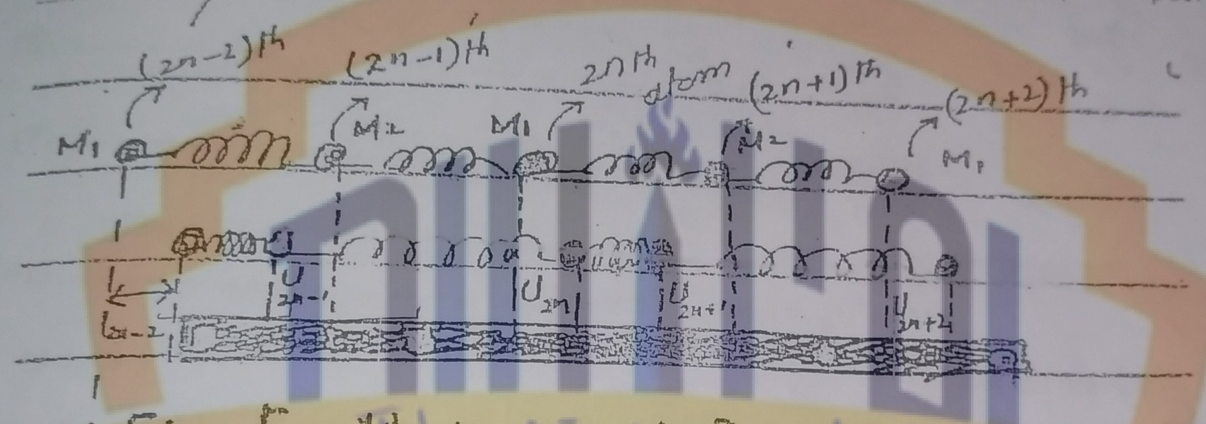


Fig: Equilibrium and Displaced position

Let $(U_{2n+1} - U_{2n})$ be the displacement of vibrations between 2n and 2n+1 atom and $(U_{2n} - U_{2n-1})$ be the displacement of vibrations between 2n and 2n-1 atoms.

By Newton's 2^{nd} law:

$$F = M_1\frac{d^2U_{2n}}{dt^2} \quad \ldots\ldots (1)$$

By using Hook's law $F = F_1 + F_2$

$$F = \beta\,(U_{2n+1} - U_{2n}) - \beta\,(U_{2n} - U_{2n-1})$$

$$F = \beta\,(u_{2n+1} - 2U_{2n} + U_{2n-1}) \quad \ldots\ldots (2)$$

Using Eq.(1)

$$M_1\frac{d^2U_{2n}}{dt^2} = \beta\,(u_{2n+1} - 2U_{2n} + U_{2n-1}) \quad \ldots\ldots (3)$$

Now for M_2:

Let $(U_{2n} - U_{2n-1})$ be the displacement of vibration b/w 2n and 2n-1 atoms and $(U_{2n-1} - U_{2n-2})$ be the displacement of vibration b/w 2n-1 and 2n-2 atoms.

$$F' = M_2 \frac{d^2U_{2n-1}}{dt^2} \quad \text{.......(4)}$$

By Hook's law:

$$F' = \beta\,(U_{2n} - U_{2n-1}) - \beta\,(U_{2n-1} - U_{2n-2})$$

Using Eq.(4)

$$M_2 \frac{d^2U_{2n-1}}{dt^2} = \beta\,(U_{2n} - U_{2n-1} - U_{2n-1} + U_{2n-2})$$

$$M_2 \frac{d^2U_{2n-1}}{dt^2} = \beta\,(U_{2n} - 2U_{2n-1} + U_{2n-2}) \quad \text{.......(5)}$$

Let the solutions of (3) and (5) are. ....(6)

$$U_{2n} = A \exp[ik(2na) - \omega t] \quad \text{.......(7)}$$

And

$$U_{2n-1} = B \exp[ik(2n-1)a - \omega t] \quad \text{.......(8)}$$

And

$$\frac{d^2U_{2n}}{dt^2} = -\,\omega^2 U_{2n} \quad \text{.......(9)}$$

$$\frac{d^2U_{2n-1}}{dt^2} = -\,\omega^2 U_{2n-1}$$

Put eq (8) in (3)

$$-\,M_1\omega^2 U_{2n} = \beta\,(u_{2n+1} - 2U_{2n} + U_{2n-1}) \quad \text{.......(10)}$$

From Eq. (6)

$$U_{2n-2} = A \exp[ik(2n-2)a - \omega t]$$

$$= A \exp[ik(2na) - ik2a - \omega t]$$

$$= A \exp[ik2na - \omega t]\ .\ \exp[-ik2a]$$

$$U_{2n-2} = U_{2n} \exp(-2ika) \quad \text{.......(11)}$$

From Eq. (7)

$$U_{2n-1} = B \exp[ik(2n-1)\,a - \omega t]$$

$$U_{2n+1} = B \exp[ik(2n+1)\,a - \omega t]$$

$$= B \exp[ik(2n-1+2)\,a - \omega t]$$

$$U_{2n+1} = B \exp\,[ik(2n-1)a + ik2a - \omega t]$$

$$= B \exp[ik(2n-1)\,a - \omega t]\,\exp(i2ka)$$

$$U_{2n+1} = U_{2n-1} \exp\,(2ika) \quad \text{.......(12)}$$

From Eq. (10)

$$-M_1\omega^2 A \exp[ik(2na - \omega t)] = \beta[U_{2n-1} \exp(2ika) - 2A\exp[ik(2na-\omega t) + B \exp[ik(2n-1)\,a - \omega t]]$$

$$-M_1\omega^2 A \exp[ik(2n-1)a - \omega t)]\, e^{ika} = \beta[U_{2n-1}\exp(2ika) - 2A\exp[ik(2n-1)a-\omega t)]\, e^{ika} + B \exp[ik(2n-$$

$1)a-\omega t]]$

$-M_1\omega^2 A\, e^{ik(2n-1)a-\omega t} e^{ika} = \beta[U_{2n-1} e^{2ika} - 2Ae^{ik(2n-1)a-\omega t} e^{ika} + Be^{ik(2n-1)a-\omega t}]$

$-M_1\omega^2 Ae^{ik(2n-1)a-\omega t} e^{ika} = \beta[Be^{ik(2n-1)a-\omega t} e^{2ika} - 2Ae^{ik(2n-1)a-\omega t}\, e^{ika} + Be^{ik(2n-1)a-\omega t}]$

$$-M_1\omega^2 A\, e^{ika} = \beta[Be^{2ika} - 2Ae^{ika} + B]$$

$$M_1\omega^2 A\, e^{ika} = \beta[2Ae^{ika} - B - Be^{2ika}]$$

$$M_1\omega^2 A\, e^{ika} - 2A\beta e^{ika} + \beta B(1+e^{2ika}) = 0$$

$$Ae^{ika}[M_1\omega^2 - 2\beta] + \beta B(1+e^{2ika}) = 0$$

$$A[M_1\omega^2 - 2\beta] + \beta B(1+e^{2ika}/e^{ika}) = 0$$

$$A[M_1\omega^2 - 2\beta] + \beta B(e^{-ika}+e^{ika}) = 0$$

$$\boxed{A(M_1\omega^2 - 2\beta) + \beta 2B\cos(ka) = 0} \quad \ldots\ldots\ldots\ldots (13)$$

Now from eq(5)

$$M_2 \frac{d^2U_{2n-1}}{dt^2} = \beta\,(U_{2n} - 2U_{2n-1} + U_{2n-2})$$

Using Eq. (9)

$$-M_2\omega^2 U_{2n-1} = \beta\,(U_{2n} - 2U_{2n-1} + U_{2n-2})$$

$$-M_2\omega^2 B\, e^{ik(2n-1)a-\omega t} = \beta[Ae^{ik2na-\omega t} - 2B\, e^{ik(2n-1)a-\omega t} + A\, e^{ik(2n-2)a-\omega t}$$

$$-M_2\omega^2 B\, e^{ik(2n-1)a-\omega t} = \beta[A\, e^{ik(2n-1)a-\omega t} e^{ika} - 2B\, e^{ik(2n-1)a-\omega t} + A\, e^{ik(2n-2)a-\omega t} e^{-ika}$$

$$-M_2\omega^2 B = \beta[Ae^{ika} - 2B + Ae^{-ika}]$$

$$M_2\omega^2 B = \beta[2B - A(e^{ika}+e^{-ika})]$$

$$M_2\omega^2 B = \beta[2B - 2ACoska]$$

$$M_2\omega^2 B - 2\beta B + 2\beta ACoska = 0$$

$$\boxed{B(M_2\omega^2 - 2\beta) + \beta 2A\cos(ka)} = 0 \quad \ldots\ldots\ldots\ldots (14)$$

Eq.(13) and Eq.(14) are homogenous simultaneous equations in the unknowns A & B. A non-trivial solution exists if the determinants of the coefficients are zero.

$$\begin{array}{cc} A & B \end{array}$$
$$\begin{vmatrix} 2\beta Coska & M_1\omega^2 - 2\beta \\ M_2\omega^2 - 2\beta & 2\beta Coska \end{vmatrix} = 0$$

$$4\beta^2 Cos^2 ka - (M_1\omega^2 - 2\beta)(M_2\omega^2 - 2\beta) = 0$$

$$4\beta^2 Cos^2 ka - [M_1M_2\omega^4 - 2\beta M_1\omega^2 - 2\beta M_2\omega^2 + 4\beta^2] = 0$$

non-trivial

$y+x=1$

By putting value of x we get y

Non-trivial

solution exist

$$M_1M_2\omega^4-2\beta M_1\omega^2-2\beta M_2\omega^2+4\beta^2-4\beta^2\cos^2 ka=0$$

$$\omega^4-2\beta\omega^2\left(\frac{M_1+M_2}{M_1M_2}\right)+4\beta^2\left(\frac{1-\cos^2 ka}{M_1M_2}\right)=0$$

$$\omega^4-2\beta\omega^2\left(\frac{M_1+M_2}{M_1M_2}\right)+4\beta^2\left(\frac{\sin^2 ka}{M_1M_2}\right)=0$$

$$(\omega^2)^2-2\beta\left(\frac{M_1+M_2}{M_1M_2}\right)\omega^2+4\beta^2\left(\frac{\sin^2 ka}{M_1M_2}\right)=0$$

$ax^2+bx+c=0$

This is a quadratic equation so,

$$(\omega^2)=\frac{2\beta\left(\frac{M_1+M_2}{M_1M_2}\right)\pm\sqrt{4\beta^2\frac{(M_1+M_2)^2}{(M_1M_2)^2}-[4(1)\left(\frac{4\beta^2\sin^2 ka}{M_1M_2}\right)}}{2(1)}$$

$$(\omega^2)=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)\pm\beta\sqrt{\frac{(M_1+M_2)^2}{(M_1M_2)^2}-\frac{4\sin^2 ka}{(M_1M_2)}} \quad \ldots\ldots\ldots (15)$$

Eq.(15) is the required dispersion relation between 'ω' and 'k' for one dimensional lattice. There are two values of ω corresponding to (+) and (-) values. For (+) value of ω we have:

$$\omega_+^2=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)+\beta\left[\left(\frac{M_1+M_2}{M_1M_2}\right)^2-\frac{4\sin^2 ka}{(M_1M_2)}\right]^{1/2}$$

For min value $\sin^2 ka=1$. This is the first Brillouin zone and it extends from $k=-\frac{\pi}{2a}$ and $k=+\frac{\pi}{2a}$

$\sin ka=\pm 1$

$ka=\sin^{-1}(\pm 1)$

$=\frac{\pi}{2}, -\frac{\pi}{2}$

$$\omega_+^2=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)+\beta\left[\left(\frac{M_1+M_2}{M_1M_2}\right)^2-\frac{4}{(M_1M_2)}\right]^{1/2}$$

$$\omega_+^2=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)+\beta\left[\frac{M_1^2+M_2^2+2M_1M_2-4M_1M_2}{(M_1M_2)^2}\right]^{1/2}$$

$$=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)+\beta\left[\left(\frac{M_1-M_2}{M_1M_2}\right)^2\right]^{1/2}$$

$$=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)+\beta\left(\frac{M_1-M_2}{M_1M_2}\right)$$

$$=\beta\frac{2M_1}{M_1M_2}$$

$$\boxed{\omega_+^2=\frac{2\beta}{M_2}} \qquad \text{OR} \qquad (\omega_+)_{min}=\sqrt{\frac{2\beta}{M_2}}$$

For (-) value of ω we have: یہاں sin کی value ① استعمال کی۔

max.

$$\omega_-^2=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)-\beta\left[\left(\frac{M_1+M_2}{M_1M_2}\right)^2-\frac{4}{(M_1M_2)}\right]^{1/2}$$

$$\omega_-^2=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)-\beta\left[\frac{M_1^2+M_2^2+2M_1M_2-4M_1M_2}{(M_1M_2)^2}\right]^{1/2}$$

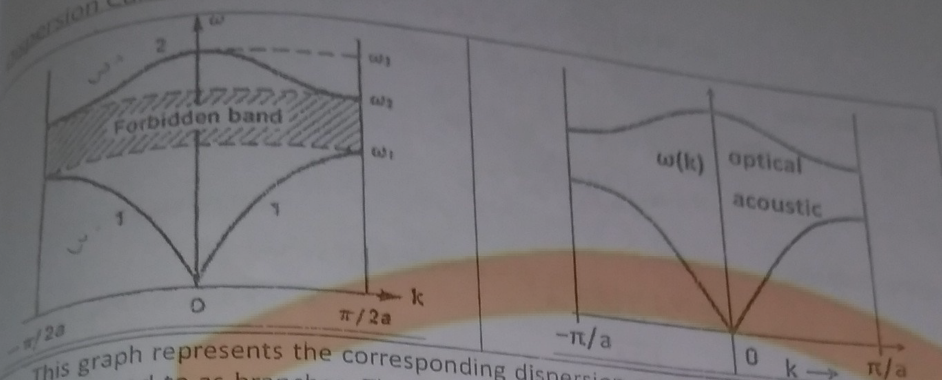
$$=\beta\left(\frac{M_1+M_2}{M_1M_2}\right)-\beta\left(\frac{M_1-M_2}{M_1M_2}\right)$$

$$=\beta\left(\frac{\cancel{M_1}+M_2-\cancel{M_1}+M_2}{M_1M_2}\right)$$

$$\omega_-^2=\frac{2\beta}{M_1}$$

یہاں پہ ہم نے ω_- کی max. کیلئے ہم نے sin کی value max value استعمال کی۔

Dispersion Curve:



This graph represents the corresponding dispersion curves in reduced zone scheme. These are referred to as branches. The lower branch of the curve corresponding the minus sign (-) in the acoustic branch, while the upper curve due to positive sign (+) is in optical branch. The former name is due to the fact that near k=0 the lower branch represents the long wavelength elastic waves with a frequency independent velocity ($\omega \propto k$), and longitudinal elastic waves are identical with sound waves. The letter is due to the fact that near k=0, the frequency of upper branch lies near infrared region of the spectrum, the corresponding wavelength is a typical wave length for infrared absorption due to molecular vibrations the frequency range b/w the top of acoustic branch and the bottom of optical branch is forbidden and the lattice cannot transmit such a wave. Therefore waves in this region are strongly attenuated by diatomic lattice which behaves as band pass mechanical filter.

The acoustic branch:

Dispersion relation corresponding to acoustic branch is given by;

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left[\left(\frac{M_1+M_2}{M_1M_2}\right)^2 - \frac{4Sin^2ka}{(M_1M_2)}\right]^{1/2}$$

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left(\frac{M_1+M_2}{M_1M_2}\right)\left[1 - \frac{4Sin^2ka}{(M_1M_2)} \times \frac{(M_1+M_2)^2}{(M_1M_2)^2}\right]^{1/2}$$

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left(\frac{M_1+M_2}{M_1M_2}\right)\left[1 - \frac{4Sin^2ka(M_1+M_2)}{(M_1M_2)^2}\right]^{1/2}$$

Since acoustic branch begins at k=0, this is mathematically defined as; $\lim_{k\to 0}(\omega_-) \to 0$

So, $Sin^2ka \approx k^2a^2$

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left(\frac{M_1+M_2}{M_1M_2}\right)\left[1 - 4k^2a^2\,\frac{(M_1M_2)}{(M_1+M_2)^2}\right]^{1/2}$$

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left(\frac{M_1+M_2}{M_1M_2}\right)\left[1 - \frac{1}{2} \times 4k^2a^2\,\frac{(M_1M_2)}{(M_1+M_2)^2}\right]$$

$$\omega_-^2 = \beta\left(\frac{M_1+M_2}{M_1M_2}\right) - \beta\left(\frac{M_1+M_2}{M_1M_2}\right) + 2\beta k^2a^2\,\frac{(M_1+M_2)}{(M_1M_2)^2}\left(\frac{M_1M_2}{M_1+M_2}\right)$$

$$\omega_-^2 = +2\beta k^2a^2\,\frac{1}{M_1M_2}$$

$$\omega_-^2 = \frac{2\beta k^2 a^2}{M_1 M_2}$$

$$\omega_- = ka\sqrt{\frac{2\beta}{M_1 M_2}}$$

Phase Velocity:

$$\lim_{k\to 0}(V_p) = \lim_{k\to 0}\left(\frac{\omega_-}{k}\right)$$

$$V = a\sqrt{\frac{2\beta}{M_1 M_2}}$$

Group Velocity:

$$\lim_{k\to 0}(V_p) = \lim_{k\to 0}\left(\frac{d\omega_-}{dk}\right)$$

$$V = a\sqrt{\frac{2\beta}{M_1 M_2}}$$

Results:

As the value of 'k' increases acoustic branch of the curve rises linearly first and then slowly decreases and eventually saturates at k=$\frac{\pi}{2a}$ and max value of allowed frequency for this mode is:

$$(\omega_-)_{max} = \sqrt{\frac{2\beta}{M_1}}$$

This is independent of lighter atoms in chain.

The optical branch:

The upper branch of the curve corresponds to the (+) sign in the optical branch and is given by,

$$\omega_+^2 = \beta\left(\frac{M_1+M_2}{M_1 M_2}\right) + \beta\left[\left(\frac{M_1+M_2}{M_1 M_2}\right)^2 - \frac{4Sin^2 ka}{(M_1 M_2)}\right]^{1/2}$$

For optical branch, wavelength is shorter and 'k' is large that's why small value of 'k' is not significant then we can put k=0, $\sin^2 ka=0$. But in acoustic branch, wavelength is larger and 'k' is shorter then small value of 'k' is is most significant and we put $k \to 0$. The above result can be written for optical branch as: $\text{Sin}^2 ka \approx k^2 a^2$ for

$$\omega_+^2 = \beta\left(\frac{M_1+M_2}{M_1 M_2}\right) + \beta\left(\frac{M_1+M_2}{M_1 M_2}\right)$$

$$\omega_+^2 = 2\beta\left(\frac{M_1+M_2}{M_1 M_2}\right)$$

$$(\omega_+)_{max} = \left[2\beta\left(\frac{M_1+M_2}{M_1 M_2}\right)\right]^{1/2}$$

Particle displacement in optical mode:

For optical mod...

long question 2

Electrostatic oR Madelung Energy

The long range interaction between ions with charge $\pm q$ is the electrostatic interaction $\pm q^2/r$, attractive between ions of opposite charge and repulsive between ions of the same charge. The main Contribution to the binding energy of ionic crystals is electrostatic and is called Madelung energy.

⇒ The sum of all Interaction for the i^{th} atoms can be written as

$$U_i = \sum_j{}' U_{ij} \quad \text{—— (1)}$$

⇒ where, the Primed summation includes all ions except $j=i$.

$$U_{ij} = \lambda \exp(-r_{ij}/\rho) \pm q^2/r_{ij} \quad \text{—— (2)}$$

⇒ where, $\pm$ represents interactions between like & opposite charges.

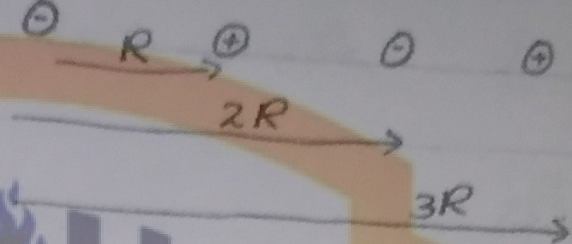
$\lambda \exp(-r_{ij}/\rho)$ = Central field repulsive Potential

λ, ρ = Empirical Parameters

⇒ It is convenient to introduce dimensionless Parameter,

$$r_{ij} = P_{ij} R$$

⇒ where R is the nearest-neighbor separation in the crystal.



As,

$$r_{ij} = P_{ij} R$$

⇒ The equation no. (2) will become

$$U_{ij} = \lambda \exp\left(-P_{ij}R/\rho\right) \pm \frac{q^2}{P_{ij}R}$$

⇒ The total lattice energy U_{tot} of a crystal composed of N molecules. As

$$U_{tot} = NU_i$$

for 2N molecules

$$U_{tot} = 2NU_i$$

⇒ Assuming ($P_{ij}=1$) for Repulsive interaction (short range)

$$U_{tot} = NU_i = N z \lambda \exp\left(-R/\rho\right) \pm \sum_j \frac{q^2}{P_{ij}R} \quad \text{—— (3)}$$

where,

⇒ Z is the number of nearest neighbors of any ion.

Replace N by 2N

Madelung Constant:

long question 3

Debye Model of specific Heat of Solids

In 1912, Debye pointed out the source of errors lies in Einstein's assumptions. According to Einstein

1- All atomic oscillators vibrate independently within a crystals.

2- The frequency of all atoms that are vibrating in a crystal is constant. It remains same for all atoms.

But according to Debye these assumptions are not justifiable since all atoms are elastically coupled to its neighbours. In this way body behaves as **elastic body**. According to Debye,

1- All atoms vibrate in a crystal and the motion of atom is dependent.

2- All vibrating atoms have different frequencies.

Day: Date:

3- When atoms vibrate about their mean position then elastic standing waves are produced. Standing waves consists of transverse and longitudinal waves.

4- The energy of standing wave is not continous but quantized.

Debye simplified the problem by considering a solid as a continously vibrating medium which give rise to **spectrum of frequencies** instead of single frequency.

The internal energy of Debye solid is given by

$$E = \int \bar{E}\, g(\omega)\, d\omega \quad \text{——— (1)}$$

where $\bar{E}$ is the average energy of harmonic oscilator with temp (T). The value of $\bar{E}$ is given as

$$\bar{E} = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{K_B T}} - 1} \quad \text{——— (2)}$$

$g(\omega)$ is the frequency distribution.

$g(\omega)d\omega$ is no. of frequencies, b/w ω and $\omega+d\omega$. Here limit is taken from $(0 \rightarrow \omega_0)$. ω_0 called upper limit of frequency.

$$g(\omega)\,d\omega = dn$$

If we consider one mole of solid then total no. of atoms in that solid is 6.02×10^{23} as avogedro's number. But if we consider no. of normal vibrations in 3D crystal lattice then no. of atoms are $3N_A$. $g(\omega)$ satifies the normalization function.

$$\int_0^{\omega_0} g(\omega)\,d\omega = 3N_A \quad \text{—— (3)}$$

Debye considered solid as a continous medium in which solution of wave eq can be written as

$$U = A\exp i(kx-\omega t)$$

Taking only spatial part

$$U = A\exp i(kx)$$

Consider 1D solid of length L since the medium is continous so that the wave function will satify the boundry conditions for period 'k'.

$$U(x=0) = U(x=L)$$

$$A\exp i(k(0)) = A\exp i(kL)$$

$$1 = \exp i(kL)$$

Day: ______

$$1 + 0i = \cos kL + i \sin kL$$

$$\cos kL = 1 \qquad \sin kL = 0$$

$$kL = 0, 2\pi, 4\pi \cdots$$

$$kL = 2\pi S \quad \because S \rightarrow \text{any integer}$$

$$\boxed{k = \frac{2\pi S}{L}} \qquad S = 0, 1, 2 \cdots$$

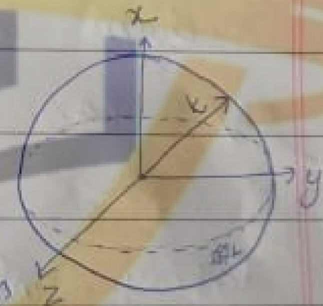
For 3D wave vector

$$k_x = \frac{2\pi S}{L}, \quad k_y = \frac{2\pi m}{L}, \quad k_z = \frac{2\pi n}{L}$$

where S, m, n are integers
Combing these we get

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

which is the eq of ~~of~~ sphere called **Debye sphere**



let n be the number of lattice points. Then we can find n by following relation

$$n = \frac{\text{Volume of the sphere}}{\text{Volume occupied by each point}}$$

$$n = \frac{4/3 \pi k^3}{(2\pi/L)^3}$$

$$n = \frac{4}{3} \pi k^3 \times \frac{L^3}{8\pi^3}$$

$$n = \frac{k^3 L^3}{6\pi^2} = \frac{k^3 V}{6\pi^2} \quad \because V = L^3$$

$$n = \frac{k^3 V}{6\pi^2}$$

$$dn = \frac{3k^2 V dk}{6\pi^2} \quad \text{—— (4)}$$

As

$$g(\omega) d\omega = dn$$

Comparing both eqs

$$g(\omega) d\omega = \frac{3k^2 V dk}{6\pi^2}$$

$$g(\omega) = \frac{3k^2 V dk}{6\pi^2 d\omega}$$

$$g(\omega) = \frac{3k^2 V}{6\pi^2 \, d\omega/dk} \quad \text{—— (5)}$$

We know that velocity of wave travelling in a continous medium is

$$V_s = f\lambda \qquad V_s \rightarrow \text{Velocity of standing wave}$$

× eq ÷ by 2π

$$V_s = 2\pi f \frac{\lambda}{2\pi}$$

$$V_s = \frac{\omega}{k} \qquad \because \omega = 2\pi f, \ k = \frac{2\pi}{\lambda}$$

$$\omega = V_s k$$

$$d\omega = V_s dk$$

$$\frac{d\omega}{dk} = V_s$$

put value in eq (5)

$$g(\omega) = \frac{3Vk^2}{6\pi^2 (V_s)}$$

$$\int_0^{\omega_0} \frac{3V\omega^2}{2\pi^2 V_s^3} d\omega = 3N_A$$

$$\frac{3V}{2\pi^2 V_s^3} \int_0^{\omega_0} \omega^2 d\omega = 3N_A$$

$$\frac{V}{2\pi^2 V_s^3} \left| \frac{\omega^3}{3} \right|_0^{\omega_0} = N_A$$

$$\frac{V\omega_0^3}{2\pi^2 V_s^3} = 3N_A$$

$$\omega_0^3 = \frac{3N_A (2\pi^2 V_s^3)}{V}$$

$$\omega_0^3 = \frac{6N_A \pi^2 V_s^3}{V} \quad \text{—— (7)}$$

Using eq (6)

$$E = \int_0^{\omega_0} \bar{E}\, g(\omega)\, d\omega$$

$$\bar{E} = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1}$$

$$E = \int_0^{\omega_0} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1} g(\omega) d\omega$$

$$E = \int_0^{\omega_0} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_BT}} - 1} \left(\frac{3V\omega^2}{2\pi^2 V_s^3} \right) d\omega$$

$$E = \int_0^{\omega_0} \frac{3\hbar V\omega^3}{2\pi^2 V_s^3 (e^{\frac{\hbar\omega}{k_BT}} - 1)} d\omega$$

$$E = \frac{3\hbar V}{2\pi^2 V_s^3} \int^{\omega_0} \frac{\omega^3}{e^{\frac{\hbar\omega}{k_BT}} - 1} d\omega$$

Day:

let $\quad x = \frac{\hbar\omega}{k_B T} \quad , \quad x_D = \frac{\hbar\omega_o}{k_B T}$

$$\omega_o = \frac{k_B T x_D}{\hbar}$$

$$dx = \frac{\hbar}{k_B T}\, d\omega$$

$$d\omega = \frac{k_B T\, dx}{\hbar} \quad , \quad \omega_o = \frac{k\,\theta_D}{\hbar} \quad \because \theta_D = T x_D$$

$\theta_D \rightarrow$ Debye Temp

$$E = \frac{3\hbar V}{2\pi^2 V_s^3} \int_0^{x_D} \frac{\left(\frac{kTx}{\hbar}\right)^3 kT\, dx}{(e^x - 1)\,\hbar}$$

$x_D \rightarrow$ Debye Displacement

$$E = \frac{3\hbar V (3N_A)}{2\pi^2 V_s^3 (3N_A)\hbar} \cdot \frac{k^3 T^3}{\hbar^3} kT \int_0^{x_D} \frac{x^3}{e^x - 1}\, dx$$

$$E = \frac{9VN_A}{6\pi^2 V_s^3 N_A} \left(\frac{kT}{\hbar}\right)^3 kT \int_0^{x_D} \frac{x^3}{e^x - 1}\, dx$$

$$E = \frac{9 N_A kT}{\frac{6\pi^2 V_s^3 N_A}{V}} \left(\frac{kT}{\hbar}\right)^3 \int_0^{x_D} \frac{x^3}{e^x - 1}\, dx$$

$\because$ $N_A k \rightarrow$ constant So we can write $\quad R = N_A k$

$$E = \frac{9N_A kT}{\omega_o^3} \left(\frac{kT}{\hbar}\right)^3 \int_0^{x_D} \frac{x^3}{e^x - 1}\, dx$$

$$\because \quad \omega_o^3 = \frac{6\pi^2 V_s^3 N_A}{V}$$

So eq becomes

$$E = \frac{9RT}{\omega_o^3} \left(\frac{kT}{\hbar}\right)^3 \int_0^{x_D} \frac{x^3}{e^x - 1}\, dx \quad \text{--- (8)}$$

This is the value of internal

Now we will calculate the value of C_v for different temperatures.

Case I

(For High Temperature ($T >> \theta_D$))

As $x_D = \frac{\theta_D}{T}$

For higher temperature x_D should be less than 1 $\quad x_D << 1$

From eq ⑧

$$e^x = 1 + x + x^2 + x^3 \cdots$$

$$e^x - 1 = x + x^2 + x^3 + \cdots$$

$e^x - 1 = x$ + neglecting high terms

$$E = \frac{9RT}{\omega_0^3}\left(\frac{k_B T}{\hbar}\right)^3 \int_0^{x_D} \frac{x^3}{x}\, dx$$

$$E = \frac{9RT}{\omega_0^3}\left(\frac{k_B T}{\hbar}\right)^3 \left(\frac{x_D^3}{3}\right)$$

$$E = \frac{\cancel{9}RT}{\cancel{\omega_0^3}}\left(\frac{\cancel{k_B T}}{\cancel{\hbar}}\right)^3 \frac{1}{3}\left(\frac{\hbar\omega_0}{k_B T}\right)^3 \quad \because x_D = \frac{\hbar\omega_0}{k_B T}$$

$$E = 3RT$$

Specific heat of solid is given as

$$C_v = \left(\frac{\partial E}{\partial t}\right)_v$$

Day: Date:

$$C_v = \frac{\partial}{\partial T}(3RT)$$

$$\boxed{C_v = 3R}$$ $\because R = 8.314\, J/mol\, K$

$R \rightarrow$ General Gas constant

$$\boxed{C_v = 5.96\, cal/mol\, K}$$

This result match both classical model and Einstein model.

Case II

(For High Temperature $\theta_D \gg T$)

when $\frac{\theta_D}{T}$ $\theta_D \gg T$ then

$$\frac{\theta_D}{T} \gg T \Rightarrow x_D \gg 1 \text{ or } x \rightarrow \infty$$

then

$$E = \frac{9RT}{\omega_D^3}\left(\frac{kT}{\hbar}\right)^3 \int_0^\infty \frac{x^3}{e^x - 1}\, dx$$

$$E = \frac{9RT}{\omega_D^3}\left(\frac{kT}{\hbar}\right)^3 \left(\frac{\pi^4}{15}\right) \quad \because \int_0^\infty \frac{x^3}{e^x - 1}\, dx = \frac{\pi^4}{15}$$

$$E = \frac{9RT}{\left(\frac{k_B \theta_D}{\hbar}\right)^3}\left(\frac{kT}{\hbar}\right)^3 \frac{\pi^4}{15} \quad \because \omega_D = \frac{k_B \theta_D}{\hbar}$$

Now

$$E = \frac{3R\pi^4 T^4}{5\theta_D^3}$$

As $C_v = \left(\frac{\partial E}{\partial t}\right)_v$

$$C_v = \frac{3R}{5\theta_D^3}\pi^4(4T^3)$$

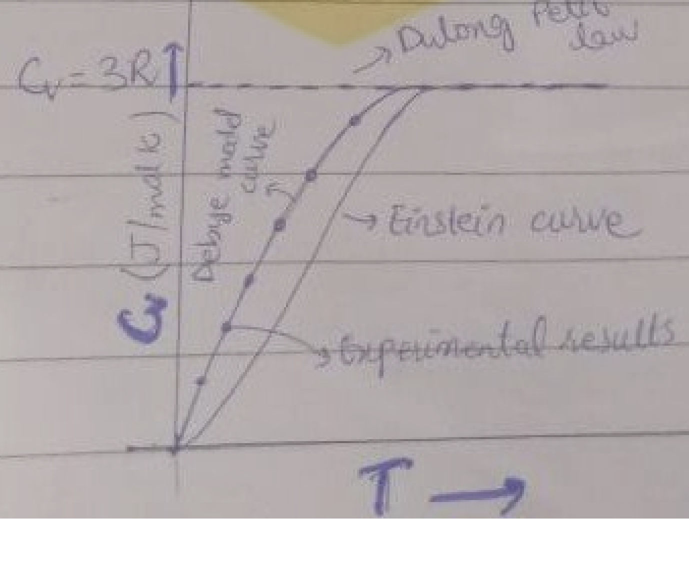
$$C_v = \frac{12R\pi^4T^3}{5\theta_D^3}$$

$$\boxed{C_v \propto T^3}$$

This is famous expression known as **Debye T^3 law** for specific heat at low temperature. This is the exact result that matches experimental results.

Graphical Explaination:-

Let us plot a graph that shows the results of measured date, Einstein model and classical values.



→ Graph shows that Dulong Petit law gives us result $C_v = 3R$ which is ~~(not)~~ constant ~~valid~~ at all temperatures. It tells us that C_v remains constant.

→ From Einstein curve, we see that C_v is valid for high temperature but its value decreases exponentially for low temperature. At low temperature, we don't get matching results of experimental data.

→ From Debye curve, we conclude that both results of high temperature and low temperature matches the experimental results. So this law is completely valid. The dots from Debye curve shows the experimental data and line shows Debye curve. Hence Debye model gives correct results for both high and low temperature.